higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MATHEMATICS N4

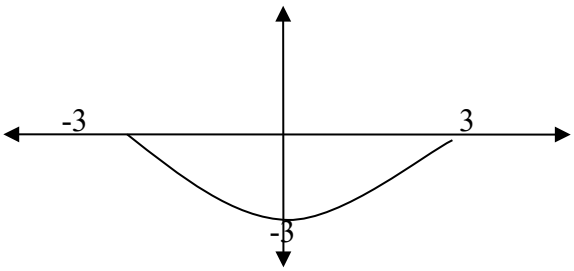
18 NOVEMBER 2019

This marking guideline consists of 9 pages.

Copyright reserved

QUESTION 1

1.1 1.1.1

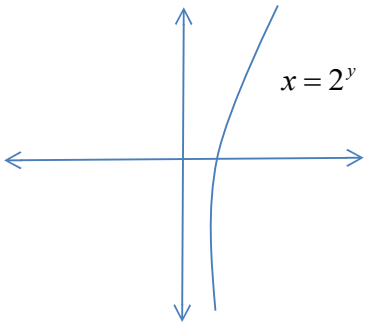


graph✓
axis✓
shape✓

(3)

1.1.2 Function (1)

1.2 1.2.1



graph√
axis√√
shape√

(4)

1.2.2 None (1)

1.2.3 Yes (1)

1.3

$$x^2+2x=-2$$
$$x^2+2x+2=0$$
$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$
$$¿\frac{-2\pm\sqrt{4-8}}{2}$$
$$¿\frac{-2\pm\sqrt{4}.\sqrt{-1}}{2}$$
$$¿\frac{-2\pm 2i}{2}$$
$$¿-1\pm i$$ ✓✓✓✓

(4)

1.4 $$\frac{\left[8\left(\cos 30^0 + j\sin 30^0\right)\right]^3}{\left(5\,cis\,45^0\right)^3}$$

$$= \frac{\left(8\angle 30^\circ\right)^3}{\left(5\angle 25^\circ\right)^3} \checkmark$$

$$= \frac{512\angle 90^\circ}{125\angle 75^\circ} \checkmark$$

$$= 4,096\angle 90^\circ - 75^\circ \checkmark$$

$$= 3,956 + j\,1,060 \quad \checkmark$$

(4)

1.5 $$j^3$$

$$= j^2 . j \quad \checkmark$$

$$= (-1).j$$

$$= -j \quad \checkmark$$

(2)
[20]

QUESTION 2

2.1 $$\sin 75^0 = \sin\left(45^0 + 30^0\right)$$

$$= \sin 45^0 . \cos 30^0 + \sin 30^0 . \cos 45^0$$

$$= \frac{1}{\sqrt{2}} . \frac{\sqrt{3}}{2} + \frac{1}{2} . \frac{1}{\sqrt{2}} \checkmark$$

$$= \frac{\sqrt{3}}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} \checkmark$$

$$= \frac{\sqrt{3}+1}{2\sqrt{2}} \checkmark$$

(3)

2.2 $$\tan\left(270^0 - x\right)$$

$$= \frac{\sin\left(270^0 - x\right)}{\cos\left(270^0 - x\right)}$$

$$= \frac{\sin 270^0 . \cos x + \cos 270^0 . \sin x}{\cos 270^0 . \cos x + \sin 270^0 . \sin x} \quad \checkmark$$

$$= \frac{-1.\cos x + 0.\sin x}{0.\cos x + (-1).\sin x} \checkmark$$

$$= \frac{-\cos x}{-\sin x}$$

$$= \cot x \checkmark$$

(3)

2.3 $5\tan^2 x-11\tan x+6=0$

$(5\tan x-6)(\tan x-1)=0$ ✓

$\tan x=\frac{6}{5}$ *or* $\tan x=1$ ✓

$\tan x=1,2$ or $x=45^0$ ✓

$=50,194$ or 45^0

$=230,194$ or 225^0 ✓✓

GS=$\{45^0;50,194^0;225^0;230,194^0\}$ ✓ (6)

2.4 $=\frac{\sin 60^0.\frac{-1}{\cos 30^0}+\tan 45^0}{-\cos 60^0+\sin 30^0}$ √ $=\frac{-\sin 60.-sec\,30+\tan 45}{\cos 60+\sin 30}$ √

$=\frac{\frac{\sqrt{3}}{2}.\frac{-2}{\sqrt{3}}+1}{\frac{-\sqrt{3}}{2}+\frac{1}{2}}$ √ $=\frac{\sqrt{3}}{2}.\frac{2}{\sqrt{3}}+1\div\frac{1}{2}+\frac{1}{2}$ √

$=0$ √ 1+1

2√ (3)

2.5 $\frac{\cos x}{1-\cos x}+\frac{\cos x}{1+\cos x}=2\cot x.\cos ecx$

LHS= $\frac{\frac{\cos x}{1-\cos x}+\frac{\cos x}{1+\cos x}}{\frac{\cos x(1+\cos x)+\cos x(1-\cos x)}{(1-\cos x)(1+\cos x)}}$ ✓

$=\frac{\cos x+\cos^2 x+\cos x-\cos^2 x}{(1-\cos^2 x)}$ ✓✓

$=\frac{2\cos x}{\sin^2 x}$ ✓

$=2\cot x.\cos ecx$ ✓ (5)

QUESTION 3

3.1 3.1.1 $y=x^3-3x$

i. For TP *let* y' = 0

$3x^2-3=0$

$x^2=1$

$x=\pm 1$ ✓

ii. For *y*-values: if $x = 1$

$y=(1)^3-3(1)$

$=-2$ ✓

If x = -1

$y=(-1)^3-3(-1)$

$=2$ ✓

$\frac{d^2y}{dx^2}=6x$

$=6(-1)$

$=-6$ maximum✓

$\frac{d^2y}{dx^2}=6x$

$=6(1)$

$=6$ minimum✓

$TP_1=(-1;2)$ maximum ✓

$TP_2=(1;-2)$ minimum ✓ (7)

3.1.2 $\frac{d^2y}{dx^2}=0$

$6x=0$

$x=0$

$y=(0)^3-3(0)$

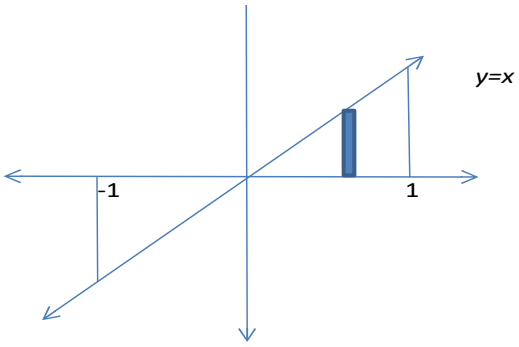
$y=0$

Points of inflection (0;0)✓✓ (2)

3.2 $y=3x.\sin 3x$

$\frac{dy}{dx}=3.\sin 3x+3x.3\cos 3x$ ✓✓✓✓ (4)

3.3



Graph✓
Area✓
Strip✓

$$A_x = \int_{-1}^{1} x\,dx$$

$$= \left[\frac{x^2}{2} + c\right]_{-1}^{1}$$

$$= \left|\left[\frac{x^2}{2} + c\right]_{-1}^{0}\right| + \left|\left[\frac{x^2}{2} + c\right]_{0}^{1}\right| \quad ✓$$

$$= \left|\frac{(0)^2}{2} + c - \left[\frac{(-1)^2}{2} + c\right]\right| + \left|\frac{(1)^2}{2} + c - \left[\frac{(0)^2}{2} + c\right]\right| \quad ✓$$

$$= \left|c - \frac{1}{2} - c\right| + \left|\frac{1}{2} + c - c\right| \quad ✓$$

$$= \left|-\frac{1}{2}\right| + \left|\frac{1}{2}\right|$$

$$= 1 \text{ units}^2 \quad ✓$$

(7)
[20]

QUESTION 4

4.1

$$y = \frac{3\sin 2x + 7}{3\sin x}$$

$$y = \frac{3.2\sin x.\cos x + 7}{3\sin x} \quad ✓$$

$$y = \frac{6\sin x.cox}{3\sin x} + \frac{7}{3\sin x} \quad ✓$$

$$y = 2\cos x + \frac{7}{3}\cos ecx$$

$$\frac{dy}{dx} = -2\sin x - \frac{7}{3}\cos ecx.\cot x \quad ✓✓$$

(4)

4.2
$$\lim_{x\to\infty}\left(\frac{x^3-3x^2+9}{x^3-7}\right)$$
$$=\frac{\frac{x^3}{x^3}-\frac{3x^2}{x^3}+\frac{9}{x^3}}{\frac{x^3}{x^3}-\frac{7}{x^3}} \quad \checkmark$$
$$=\frac{1-\frac{3}{\infty}+\frac{9}{\infty^3}}{1-\frac{7}{\infty^3}} \quad \checkmark$$
$$=1 \quad \checkmark$$
(3)

4.3
$$\int\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)dx$$
$$=\int x^{\frac{1}{2}}dx+\int x^{\frac{-1}{2}}dx \quad \checkmark$$
$$=\frac{x^{\frac{3}{2}}}{\frac{3}{2}}+\frac{x^{\frac{1}{2}}}{\frac{1}{2}}+c \quad \checkmark$$
$$=\frac{2}{3}\sqrt{x^3}+2\sqrt{x}+c \quad \checkmark$$
(3)

4.4
$$\int_1^3\left(\frac{5}{x}+6\right)dx$$
$$=\left[5\ln x+6x+c\right]_1^3 \quad \checkmark$$
$$=5\ln(3)+6(3)+c-(5\ln(1)+6(1))+c \checkmark$$
$$=5,493+18-6$$
$$=17{,}493 \quad \checkmark$$
(3)

4.5
$$\int\left(3\cos\pi x-5\sqrt{x^3}+\frac{1}{\sqrt{x}}-2^{-6x}-5e^{-4x}+7y\right)dx$$
$$=\frac{-3}{\pi}\sin\pi x-2x^{\frac{5}{2}}+2x^{\frac{1}{2}}+\frac{2^{-6x}}{6\ln x}+\frac{5e^{-4x}}{4\ln e}+7yx+c$$
✓ ✓ ✓ ✓ ✓ ✓ ✓
(7)
[20]

QUESTION 5

5.1
$l=2b$.....1 ✓
$l\times b=128$ .....2 ✓
$2b\times b=128$
$2b^2=128$ ✓
$b^2=64$
$b=8$ ✓
$l=2.8$
$l=16$ ✓ (5)

5.2
$2^{-x+2}=6^{2x-3}$
$\ln 2(-x+2)=\ln 6(2x-3)$ ✓
$-0,693x+1,386=3,584-5,375$ ✓
$6,068x=6,761$
$x=1,114$ ✓ (3)

5.3
$q=q\left(1-e^{\frac{-t}{r}}\right)_0$
$-e^{\frac{-t}{r}}=\frac{q}{q_0}-1$ ✓
$e^{\frac{-t}{r}}=1-\frac{q}{q_0}$
$\ln e^{\frac{-t}{r}}=\ln\left(1-\frac{q}{q_0}\right)$
$-\frac{t}{r}\ln e=\ln\left(1-\frac{q}{q_0}\right)$ ✓
$t=-r\ln\left(1-\frac{q}{q_0}\right)$ ✓ (3)

5.4
$$\begin{vmatrix} 5 & -6 & 3 \\ 2 & -3 & 2 \\ 3 & -7 & 5 \end{vmatrix}$$
$$D = 5\begin{vmatrix} -3 & 2 \\ -7 & 5 \end{vmatrix} + 6\begin{vmatrix} 2 & 2 \\ 3 & 5 \end{vmatrix} + 3\begin{vmatrix} 2 & -3 \\ 3 & -7 \end{vmatrix} \checkmark$$
$$= -5 + 24 - 15$$
$$= 4 \checkmark$$
$$D_a = \begin{vmatrix} -9 & -6 & 3 \\ -5 & -3 & 2 \\ -16 & -7 & 5 \end{vmatrix}$$
$$= -9\begin{vmatrix} -3 & 2 \\ -7 & 5 \end{vmatrix} + 6\begin{vmatrix} -5 & 2 \\ -16 & 5 \end{vmatrix} + 3\begin{vmatrix} -5 & -3 \\ -16 & -7 \end{vmatrix} \checkmark$$
$$= 9 + 42 - 39$$
$$= 12 \checkmark$$
$$a = \frac{D_a}{D}$$
$$= \frac{12}{4}$$
$$= 3 \checkmark$$ (5)

5.5 5.5.1
$$\begin{vmatrix} -6 & 2 \\ 1 & 1 \end{vmatrix} \checkmark$$
$$= -6 - 2$$
$$= -8 \checkmark$$

5.5.2
$$(-1)^{i+j}\begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} \checkmark$$
$$= 2 + 1$$
$$= 3 \checkmark$$

(2 × 2) (4)
[20]

TOTAL: 100

Copyright reserved